

Refund Hunter: the demo guide

Everything you need to set it up, understand it, record the video, and submit. Step by step, with the screens you will see.

1. How it works, in one minute

Refund Hunter is an agent that reads your order emails, finds the refunds stores owe you (price drops inside a price-adjustment window, late deliveries), and files the claim after you tap **Yes**. If it finds nothing, you never hear from it.



- **Two agents**, built with the Strands Agents SDK, run on Claude (Amazon Bedrock).
- **The Hunter has five tools**. One of them, `file_claim`, stops the agent with a Strands **interrupt** and waits for your answer. Your answer resumes the exact same agent, even hours later, because the session is saved.
- **It runs by itself once a day** on Amazon Bedrock AgentCore, woken by an EventBridge schedule. Memory is in DynamoDB, paused sessions in S3.
- **The demo inbox is fake**. A made-up person, made-up orders. No SMTP is configured, so claim emails stay in the "Emails the agent sent" list instead of going to a real store.

2. Setup

Start the dashboard. Open a terminal in the project folder and run:

```
cd refund-hunter\backend
.venv\Scripts\refundhunter serve
```

Then open **http://127.0.0.1:8000** in Chrome. Leave the terminal open while you record.

Fresh start: press **Reset demo** in the dashboard (top right) before recording. It wipes purchases, claims, and answers.

Connecting your own AI key and inbox (for real use, not needed for the video)

Click **⚙ Settings** at the top right. The form asks for four things: the AI (an Anthropic API key, or AWS Bedrock), your inbox (the demo inbox, or your Gmail with an App Password), whether the agent may send claim emails, and your name. **Test connection** checks the inbox login and the AI in one go. Everything is saved to a private file on this computer and never uploaded.

The screenshot shows the 'Refund Hunter' application interface. At the top, there's a header with the title 'Refund Hunter' and a subtitle 'Sam Rivera · local agent · dry-run outbound'. On the right, there are buttons for 'Run today's check', 'Reset demo', and '⚙ Settings'. Below the header, a blue banner states: 'You're on the **demo inbox**. To hunt refunds in your own mail, add your AI key and Gmail in Settings. [Open Settings →](#)'. The main content area is divided into several sections: 'MONEY FOUND' showing '\$0.00', 'CLAIMS FILED', 'NEEDS YOU' with a 'Needs you 0' indicator and the text 'Nothing needs you. The agent is busy.', 'PURCHASES WATCHED', 'Claims' with 'No claims yet.', 'Purchases' with 'No purchases yet. Run today's check', 'Emails the agent sent' with '0' and 'Nothing sent yet.', and 'Runs' with 'No runs yet.'. A 'Settings' modal is open in the center. It has a title 'Settings' and a note: 'Everything here stays on this computer in a private file. Nothing is uploaded anywhere.' The modal is divided into four sections: 1. THE AI THAT READS AND DECIDES, with radio buttons for 'Anthropic API key' and 'AWS Bedrock (uses your AWS login)', and input fields for 'AWS region' (us-east-1) and 'Bedrock model id' (global.anthropic.claude-sonnet-4-6); 2. YOUR INBOX, with radio buttons for 'Demo inbox (made-up receipts)' and 'My Gmail'; 3. SENDING CLAIMS, with checkboxes for 'Let the agent send claim emails from my Gmail (after I tap Yes)' and 'Safe mode: send every claim to me instead of the store, so I can read it first'; 4. YOU, with input fields for 'Your name (signs the emails)' (Sam Rivera), 'Your email (reply address)' (sam.rivera.demo@example.com), 'Follow up after (days)' (5), and 'Ignore refunds under (\$)' (3). At the bottom of the modal are buttons for 'Test connection', 'Cancel', and 'Save'.

Useful commands, if you ever want them:

COMMAND	WHAT IT DOES
refundhunter run	Same as the "Run today's check" button, in the terminal
refundhunter decisions	Lists the questions waiting for you
refundhunter reset	Wipes the demo state

On a new computer (not needed today): install uv, then in backend/ run `uv venv .venv --python 3.12, uv pip install -e ".[dev]" "strands-agents[anthropic]"`, copy `.env.example` to `.env`, and make sure AWS credentials with Bedrock access are configured.

3. The demo, screen by screen

Record your screen at 1280 px wide or more. Total target: **4 min 30 s**. The gray time stamps are where you should be.

0:00 Slide or talking head: the problem

Say: *"Stores owe you money more often than you think. Target refunds the difference if their price drops within 14 days of your order. Costco gives you 30 days. Amazon refunds the shipping you paid when it misses a guaranteed delivery date. Almost nobody collects it, because for \$22 you'd have to notice, know the rule, find the order number, write the email, and chase the reply. So the money stays with the store."*

0:40 Who it's for

Say: *"Refund Hunter is for anyone who shops online. It reads your receipts, checks every store's rules and prices in the background, and asks you exactly one question when it finds money. If there's nothing to claim, you never hear from it. Built with the Strands Agents SDK, running on Amazon Bedrock AgentCore."*

1:10 Screen 1: the empty dashboard

Do: show the dashboard after Reset. Point at the four tiles and the empty "Needs you" box.

Refund Hunter

Sam Rivera · local agent · dry-run outbox

Run today's check

Reset demo

MONEY FOUND

\$0.00

CLAIMS FILED

\$0.00

NEEDS YOU

0

PURCHASES WATCHED

0

Needs you

0

Nothing needs you. The agent will ask here when it finds money and needs a yes.

Claims

No claims yet.

Purchases

0

No purchases yet. Run today's check to read the inbox.

Emails the agent sent

0

Nothing sent yet.

Due

Say: "This is Sam's dashboard. Nothing yet. Sam's inbox has a dozen emails from the last three weeks: orders from Target, Costco, Amazon, Best Buy, Walmart, Home Depot, an Instacart delivery, a newsletter, and a note from a friend."

Optional: show backend/fixtures/inbox.json for five seconds so people see the emails are made up.

1:30 Screen 2: press "Run today's check"

Do: press the black button. It takes 30 to 45 seconds. Keep talking while it works.

Say: *"The first agent, the Receipt Reader, turns those emails into a list of purchases: what, where, how much, when it was promised. The second agent, the Hunter, looks up each store's policy, checks today's price, compares delivery dates, and decides."*

MONEY FOUND

\$17.99

CLAIMS FILED

\$0.00

NEEDS YOU

2

PURCHASES WATCHED

5

Needs you 2

Amazon: claim \$12.99 for late delivery on "LEGO Icons Botanical Collection Orchid 10311"?

Paid \$12.99 shipping for a guaranteed delivery of 2026-09-08; item was not delivered until 2026-09-10, two days late.

Rule: Amazon refunds shipping fees paid when a guaranteed delivery date is missed, upon request.

Yes, file it

Skip

Instacart / Sprouts: claim \$5.00 for late delivery on "Sprouts grocery order, 14 items"?

Order was promised by 6:00 PM on 2026-09-12 but was delivered at 7:48 PM, 1 hour 48 minutes after the scheduled window ended.

Rule: Instacart offers a \$5.00 fixed late-delivery credit for orders delivered after the promised window, available through Instacart support.

Yes, file it

Skip

Claims

STORE	ITEM	KIND	AMOUNT	STATUS
Instacart / Sprouts	Sprouts grocery order, 14 items Order was promised by 6:00 PM on 2026-09-12 but was delivered at 7:48 PM, 1 hour 48 minutes after the scheduled window ended.	late delivery	\$5.00	pending approval
Amazon	LEGO Icons Botanical Collection Orchid 10311 Paid \$12.99 shipping for a guaranteed delivery of 2026-09-08; item was not delivered until 2026-09-10, two days late.	late delivery	\$12.99	pending approval

Purchases 7

STORE	ITEM	PAID	ORDERED	DELIVERY	STATUS
Instacart / Sprouts	Sprouts grocery order, 14 items IC-88214-7710	\$86.40 + \$3.99 shipping	2026-09-12	promised 2026-09-12 arrived 2026-09-12 scheduled window ended 6:00 PM; delivered at 7:48 PM	open
Walmart	Instant Pot Duo 7-in-1 Electric Pressure Cooker, 6 Qt 2000418-77193	\$89.00	2026-09-08	promised 2026-09-11 arrived 2026-09-11 closed: Walmart has no price-adjustment policy and delivery was on time (promised and delivered 2026-09-11); nothing to claim.	closed
Amazon	LEGO Icons Botanical Collection Orchid 10311 113-4471820-2291001	\$49.99 + \$12.99 shipping	2026-09-06	promised 2026-09-08 arrived 2026-09-10	open
Target	Ninja Foodi 6-in-1 8-qt 2-Basket Air Fryer (DZ201) 102-5583920	\$129.99	2026-09-05	promised 2026-09-09	open
Costco	Dyson V11 Cordless Stick Vacuum, Extra Tools 5211093774	\$499.99	2026-09-02		open
Home Depot	DEWALT 20V MAX Cordless Drill/Driver Kit WG73015566	\$99.00	2026-08-29		open
Best Buy	Sony WH-1000XM5 Wireless Noise Canceling Headphones - Black 23066-005553361	\$348.00	2026-08-25	closed: Outside Best Buy's 15-day price-adjustment window (window closed 2026-09-09) and no delivery promise to check.	closed

Say: "Seven purchases. Best Buy is closed: 20 days old, outside the 15-day window. Walmart is closed: no price adjustment, delivered on time. And up here, the agent has four questions for me. It didn't send anything. Costco dropped my vacuum by \$50, 12 days into a 30-day window. Target dropped the air fryer by \$22. Amazon delivered two days after

the guaranteed date, and I paid \$12.99 for shipping. Instacart delivered an hour and 48 minutes after the window."

Sometimes the questions arrive in two waves: a couple right after the run, the rest after you answer the first one. That is normal, and it looks good on video: "it found more."

2:20 **Screen 3: tap "Yes, file it"**

Do: tap **Yes, file it** on the Costco question (\$50). Wait about 5 seconds. Scroll down to "Emails the agent sent" and open the email.

Say: *"This is the part that matters. Filing a claim is a Strands interrupt: the agent stopped in the middle of its job and waited for me. Now it resumes, writes the email with the order number, the dates, the rule it relies on, and the amount, and sends it."*

Refund Hunter

Sam Rivera · local agent · dry-run outbox

Run today's check

Reset demo

MONEY FOUND

\$89.99

CLAIMS FILED

\$50.00

NEEDS YOU

3

PURCHASES WATCHED

4

Needs you 3

Target: claim \$22.00 for price drop on "Ninja Foodi 6-in-1 8-qt 2-Basket Air Fryer (DZ201)"?

Paid \$129.99 on 2026-09-05; current Target price is \$107.99, a drop of \$22.00. Today (2026-09-14) is 9 days after the order, within the 14-day adjustment window.

Rule: Target matches its own lower price within 14 days of purchase.

Yes, file it

Skip

Amazon: claim \$12.99 for late delivery on "LEGO Icons Botanical Collection Orchid 10311"?

Amazon guaranteed delivery by 2026-09-08; order was delivered on 2026-09-10, two days late. Shipping fee paid was \$12.99.

Rule: If a guaranteed delivery date is missed, Amazon refunds the shipping fees paid for that shipment on request.

Yes, file it

Skip

Instacart / Sprouts: claim \$5.00 for late delivery on "Sprouts grocery order, 14 items"?

Instacart/Sprouts order on 2026-09-12 had a scheduled delivery window ending at 6:00 PM; it was delivered at 7:48 PM, 1 hour and 48 minutes late. Instacart's fixed late-delivery credit is \$5.00.

Rule: Orders delivered after the promised window can request a late-delivery credit through Instacart support (fixed \$5.00 credit).

Yes, file it

Skip

Claims

STORE	ITEM	KIND	AMOUNT	STATUS	
Costco	Dyson V11 Cordless Stick Vacuum, Extra Tools	price drop	\$50.00	filed	Got the refund Refused
Target	Ninja Foodi 6-in-1 8-qt 2-Basket Air Fryer (DZ201)	price drop	\$22.00	pending approval	
Amazon	LEGO Icons Botanical Collection Orchid 10311	late delivery	\$12.99	pending approval	
Instacart / Sprouts	Sprouts grocery order, 14 items	late delivery	\$5.00	pending approval	

Purchases 7

STORE	ITEM	PAID	ORDERED	DELIVERY	STATUS
Instacart / Sprouts	Sprouts grocery order, 14 items IC-88214-7710	\$86.40 +\$3.99 shipping	2026-09-12	promised 2026-09-12 arrived 2026-09-12 scheduled window ended 6:00 PM; delivered at 7:48 PM	open
Walmart	Instant Pot Duo 7-in-1 Electric Pressure Cooker, 6 Qt 2000418-77193	\$89.00	2026-09-08	promised 2026-09-11 arrived 2026-09-11 closed: Walmart has no price-adjustment policy and the order was delivered on time on 2026-09-11 as promised.	closed

3:00

Screen 4: skip one, approve the rest

Do: tap **Skip** on Target (\$22). Tap **Yes** on Amazon (\$12.99) and Instacart (\$5). Point at the tiles at the top.

Say: *"Maybe I don't want to bother Target for \$22. Skip. Remembered, no email. Amazon, two days late on a guaranteed date: yes, shipping refunded. Instacart: yes. Money found, claims filed. Every email is here, and I can read it."*

MONEY FOUND

\$67.99

CLAIMS FILED

\$67.99

NEEDS YOU

0

PURCHASES WATCHED

2

Needs you 0

Nothing needs you. The agent will ask here when it finds money and needs a yes.

Claims

STORE	ITEM	KIND	AMOUNT	STATUS	
Instacart / Sprouts	Sprouts grocery order, 14 items Instacart/Sprouts order on 2026-09-12 had a scheduled delivery window ending at 6:00 PM; it was delivered at 7:48 PM, 1 hour and 48 minutes late. Instacart's fixed late-delivery credit is \$5.00.	late delivery	\$5.00	filed	Got the refund
					Refused
Amazon	LEGO Icons Botanical Collection Orchid 10311 Amazon guaranteed delivery by 2026-09-08; order was delivered on 2026-09-10, two days late. Shipping fee paid was \$12.99.	late delivery	\$12.99	filed	Got the refund
					Refused
Target	Ninja Foodi 6-in-1 8-qt 2-Basket Air Fryer (DZ201) Paid \$129.99 on 2026-09-05; current Target price is \$107.99, a drop of \$22.00. Today (2026-09-14) is 9 days after the order, within the 14-day adjustment window.	price drop	\$22.00	skipped skipped by owner	
Costco	Dyson V11 Cordless Stick Vacuum, Extra Tools Paid \$499.99 on 2026-09-02; current Costco.com price is \$449.99, a drop of \$50.00. Today (2026-09-14) is 12 days after the order, within the 30-day adjustment window.	price drop	\$50.00	filed	Got the refund
					Refused

Purchases 7

STORE	ITEM	PAID	ORDERED	DELIVERY	STATUS
Instacart / Sprouts	Sprouts grocery order, 14 items IC-88214-7710	\$86.40 +\$3.99 shipping	2026-09-12	promised 2026-09-12 arrived 2026-09-12 scheduled window ended 6:00 PM; delivered at 7:48 PM	claimed
Walmart	Instant Pot Duo 7-in-1 Electric Pressure Cooker, 6 Qt 2000418-77193	\$89.00	2026-09-08	promised 2026-09-11 arrived 2026-09-11 closed: Walmart has no price-adjustment policy and the order was delivered on time on 2026-09-11 as promised.	closed
Amazon	LEGO Icons Botanical Collection Orchid 10311 113-4471820-2291001	\$49.99 +\$12.99 shipping	2026-09-06	promised 2026-09-08 arrived 2026-09-10	claimed
Target	Ninja Foodi 6-in-1 8-qt 2-Basket Air Fryer (DZ201) 102-5583920	\$129.99	2026-09-05	promised 2026-09-09	open
Costco	Dyson V11 Cordless Stick Vacuum, Extra Tools 5211093774	\$499.99	2026-09-02		claimed
Home Depot	DEWALT 20V MAX Cordless Drill/Driver Kit WG73015566	\$99.00	2026-08-29		open
Best Buy	Sony WH-1000XM5 Wireless Noise Canceling Headphones - Black BBY01-806623014	\$348.00	2026-08-25	closed: Outside Best Buy's 15-day price-adjustment window (ordered 2026-08-25, now 20 days ago); no delivery issue to check.	closed

Emails the agent sent 3

▼ Late delivery refund request, order IC-88214-7710 → support@instacart.example outbox (dry run)

Hello Instacart / Sprouts team,

My order IC-88214-7710 (Sprouts grocery order, 14 items) had a promised delivery of September 12, 2026 and arrived September 12, 2026.

Orders delivered after the promised window can request a late-delivery credit through Instacart support (fixed \$5.00 credit).

Could you please refund \$5.00 to my original payment method?

Thank you,
Sam Rivera
sam.rivera.demo@example.com

▼ Late delivery refund request, order 113-4471820-2291001 → delivery.guarantee@amazon.example outbox (dry run)

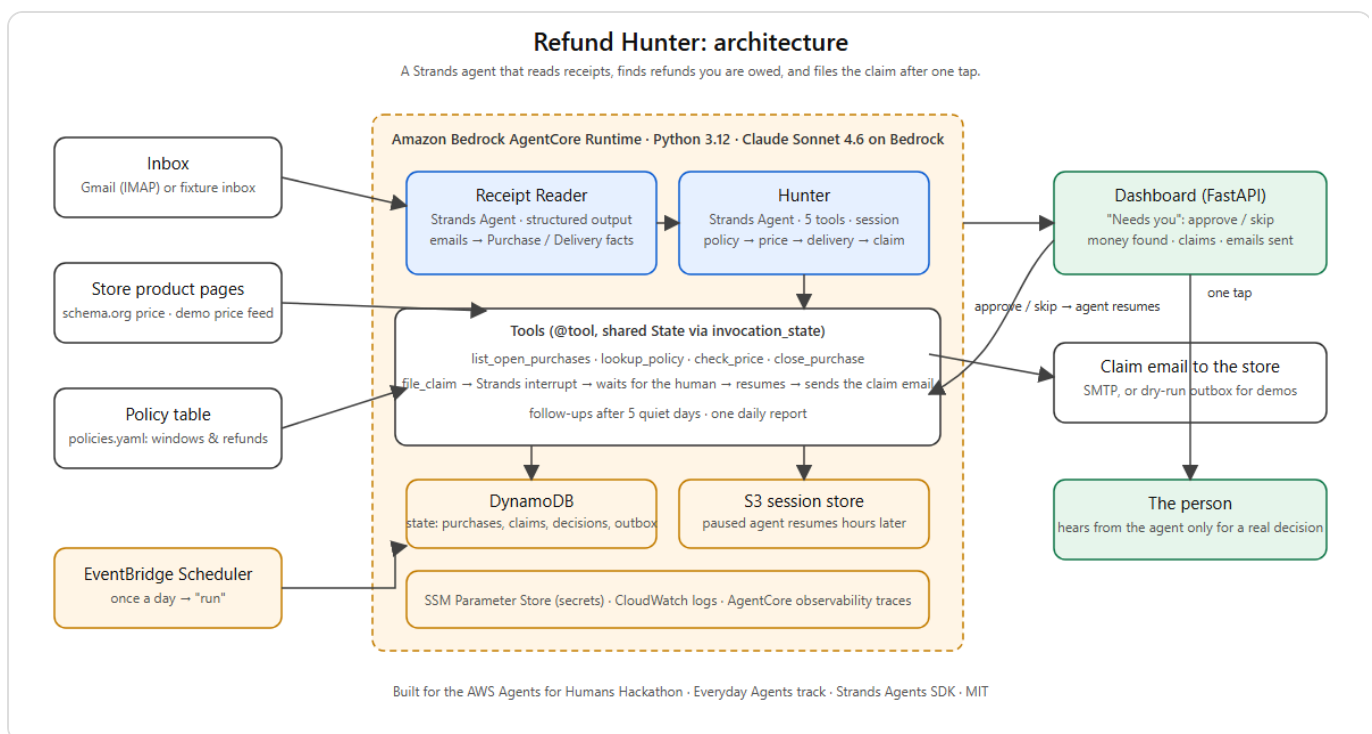
3:40 Background and follow-up

Say: "Every morning at eight, EventBridge wakes the agent on AgentCore. New receipts get read, prices get re-checked while purchases are still inside their windows, and any claim the store hasn't answered in five days gets one polite follow-up. I only see this page when there's a decision."

Do: show the AWS console for 10 seconds: Amazon Bedrock → AgentCore → Runtimes → **RefundHunter_Hunter** (status READY). If you have time, EventBridge Scheduler → **refundhunter-daily**.

4:00 Architecture

Do: show docs/architecture.png full screen.



Say: "Two Strands agents. The Reader uses structured output. The Hunter has five tools, and one of them raises the interrupt. State in DynamoDB, the paused agent in S3, Claude on Bedrock, EventBridge for the schedule. Everything in the demo is synthetic, and the same code reads a real Gmail inbox over IMAP."

4:30 Close

Say: "Refund Hunter: the agent that collects the small money you're owed, and only talks to you when it needs a yes. It's open source, MIT, on GitHub."

4. Recording tips

- Use a clean Chrome window: no bookmarks bar, no other tabs visible. Zoom 100%.
- Do one full dry run first, then Reset demo and record.
- If the run takes long, cut the wait in editing. Judges want under 5 minutes.
- Windows: **Win + Alt + R** records the screen (Xbox Game Bar), or use OBS.
- Upload to YouTube as **Public**. Title: "Refund Hunter - Agents for Humans Hackathon (Strands Agents + AgentCore)".

5. Submit on Devpost

1. Go to <https://agentsforhumans.devpost.com> and start a submission.
2. Project name **Refund Hunter**. Track **Everyday Agents**.
3. Description: copy the text from `docs/devpost.md`.
4. Repo: **<https://github.com/fakemovement/refund-hunter>** (public, MIT shown in About).
5. Video: your YouTube link.
6. Images: upload `docs/architecture.png` and `docs/dashboard.png`.
7. AWS Builder ID email: yours.
8. Pre-existing code: "Deployment scaffold (AgentCore CDK project) reused from an earlier project of mine; all agent code written during the submission period."
9. Submit before **1 PM** Pacific. The hard deadline is 5 PM.